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SUMMARY

Backend engineer (3+ years, Python and Java) now working on production LLM systems, mostly evaluation, guardrails, and monitoring. Previously built KYC services for Citi's 48M-user platform; now on generative AI at Rokt.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, SQL
Backend & Data: Spring Boot, FastAPI, Kafka, Redis, PostgreSQL, REST microservices, React, Next.js
Generative AI: LLM evaluation & guardrails, structured outputs, hybrid RAG (vector + BM25), LangGraph, MCP, LangChain
Cloud & Reliability: AWS (EKS, EC2, S3), Docker, Kubernetes, Terraform, OpenTelemetry, Prometheus, Jenkins CI/CD
AI Dev Tools: Claude Code, Cursor

EXPERIENCE

Rokt (Contract via Axitem) **Aug 2025 – Present**
Software Engineer / AI Engineer *New York, NY*
Customer data platform (mParticle CDP) | Python, FastAPI, Java, React/TypeScript, Redis, OpenTelemetry, AWS

- Created a React/TypeScript tool to trace mParticle CDP profile merges, caching Spring Boot API data via React Query; cut triage **30%**.
- Cut p95 latency of real-time decisioning APIs **29% (185 ms to 132 ms) at ~50k RPS** by replacing sequential network calls with parallel asyncio retrieval and adding a Redis caching layer.
- Shipped consent-aware data routing to **300+ partner integrations**, streaming identity-resolved attributes through Java/Spring Boot APIs with GDPR consent and CCPA opt-out enforced before data reached decisioning.
- On the on-call rotation, added OpenTelemetry tracing, Prometheus metrics, and Chronosphere alerts that helped the team cut incident detection from **20 to 8 minutes** and resolution from **60 to 25**.
- Developed Python LLM evaluation and guardrails for Rokt's generative ad-content engine: JSON schema validation and automated quality filters on 1.5M headlines/day, **cutting manual rewrites 45%**.
- Wrote Python and Java services test-first with pytest and JUnit **(85% coverage)** and took part in design and code reviews.

Binghamton University **Jan 2025 – May 2025**
Software Engineer (Part-time, during MS) *Binghamton, NY*

- Owned an on-premises AI teaching agent serving **~200 student queries/day** with no cloud APIs, keeping student data on-prem (FERPA).
- Raised answer accuracy with hybrid RAG (LangChain, ChromaDB, Ollama), fusing vector and BM25 search via Reciprocal Rank Fusion.
- Exposed course docs as **3 MCP tools** behind a LangGraph agent with step limits and fallbacks, cutting context overhead.
- Served Llama 3.1 8B with 4-bit quantization and asyncio request batching, dropping latency **43% (1.2 s to 680 ms)**.
- Saved **~$400/month** versus cloud LLM APIs by running inference on-prem, backed by a cost-per-query analysis.

Virtusa (Client: Citi) **May 2021 – Jul 2023**
Software Engineer *Hyderabad, India*

- Delivered Java/Spring Boot services and React features for customer onboarding and KYC on a **48M-user banking platform**.
- Integrated Onfido identity verification with Kafka-streamed status updates, cutting verification time by **about two-thirds**, and autoscaled the Kafka consumers on AWS EKS for peak volume.
- Connected **15+ third-party REST APIs** to the payments and banking module and provisioned AWS (EC2, S3) with Terraform.
- Automated Jenkins CI/CD build pipelines, which cut post-deployment issues by **30%**.

Virtusa **Nov 2020 – Mar 2021**
Software Engineer Intern *Hyderabad, India*

- Wrote a reusable Spring Boot/MVC REST layer for data retrieval, so developers could add new endpoints faster.
- Implemented LDAP/JWT authentication with **90%** JUnit test coverage.

PROJECTS

Source-Grounded RAG Chatbot (Freelance, NDA) | Python, LangChain, Pinecone, FastAPI, AWS

- Built a chatbot that answers from a website's own content, with Pinecone embeddings and FastAPI on auto-scaling Elastic Beanstalk.
- Brought hallucination rate from **35% to under 8%** by requiring every answer to cite the exact chunk it came from.

HireFast – Multi-Tenant Candidate Search | TypeScript, Next.js, PostgreSQL, Stripe, Claude API

- Designed the PostgreSQL schema and PL/pgSQL search over **5,000 profiles**, with the Claude API turning free-text recruiter queries into GIN-indexed lookups and index-range paging.
- Isolated tenants with row-level security and deny-by-default grants; made the free-tier limit **race-proof with one atomic conditional UPDATE** (concurrency test: 10 requests, 1 success, 9 rejected).
- Wrote a Stripe webhook handler with HMAC verification, idempotent event-ID processing, and an out-of-order guard.

EDUCATION & CERTIFICATIONS

State University of New York at Binghamton, Master of Science in Computer Science Aug 2023 – May 2025
Malla Reddy College of Engineering and Technology, Bachelor of Technology in Information Technology
AWS Certified AI Practitioner | AWS Certified Cloud Practitioner